# Exact Geodesic Recovery of the Kahneman–Tversky Value Function via Metric Inversion in Predictive Interaction Field Theory

An Extension of $\mathrm{PIFT}_\gamma$: From Asymmetric Curvature Amplification to Isometric Embedding


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**Abstract**

Predictive Interaction Field Theory (PIFT) models behavioral dynamics as geodesic flow on a Riemannian manifold whose metric is shaped by the Expected Free Energy $\mathcal{G}$. A preceding paper ($\mathrm{PIFT}_\gamma$) demonstrated that introducing an asymmetric geometric response function $\Psi(\Delta)$ into the metric $g_{\mu\nu}$ generates loss aversion whose Christoffel ratio reproduces the empirical Kahneman–Tversky (KT) ratio $\lambda = 2.25$, but the resulting geodesic value function $L(\Delta)$ diverged visually from the KT value function $V_{\mathrm{KT}}(\Delta)$ due to a fundamental power-law mismatch introduced by path integration.

This paper identifies the root cause of that mismatch and derives an *exact solution*: the unique metric $g_{11}(\Delta)$ for which geodesic length recovers the KT value function *identically*. The solution is obtained by inverting the arc-length functional: $g_{11}(\Delta) = \left[dV_{\mathrm{KT}}/d\Delta\right]^2$. This yields a power-law metric with an integrable singularity at $\Delta = 0$ (the gain/loss boundary) and an intrinsic loss-aversion ratio of $\lambda^2 = 5.0625$ at the metric level—the square of the behavioral KT ratio. We prove that this singularity is integrable for all $\alpha \in (0, 1)$, confirm numerical convergence at machine-precision RMSE, and discuss the implications for the PIFT field equations and the interpretation of $\nabla_{I_S}\gamma \neq 0$. The result constitutes a geometrically rigorous, regularization-free embedding of Prospect Theory into Riemannian interaction geometry.



**Keywords:** Predictive Interaction Field Theory, Prospect Theory, Kahneman–Tversky, loss aversion, Riemannian geodesic, metric inversion, Expected Free Energy, integrable singularity.

# Contents

# 1 Introduction

The Kahneman–Tversky (KT) value function [1, 2]

$$V_{\mathrm{KT}}(\Delta) = \begin{cases} \Delta^{\alpha}, & \Delta \geq 0, \\ -\lambda\,(-\Delta)^{\alpha}, & \Delta < 0, \end{cases} \tag{1}$$

with empirical constants $\lambda = 2.25$ and $\alpha = 0.88$, is the canonical description of human decision asymmetry. In Prospect Theory this function is inserted phenomenologically as a utility representation.

Predictive Interaction Field Theory (PIFT), developed in a series of companion papers [3, 4, 5], proposes instead that behavioral dynamics are geodesic trajectories on a Riemannian manifold $(\mathcal{M}, g_{\mu\nu})$ whose curvature is sourced by the Expected Free Energy (EFE) $\mathcal{G} \approx A + R$ (Ambiguity plus Relative Risk). The paper PIFT$_{\gamma}$ [6] showed that an asymmetric perturbative deformation

$$g_{\mu\nu}(\Delta) = g^{(0)}_{\mu\nu} + \epsilon\,\Psi(\Delta)\,Q_{\mu\nu}, \qquad \Psi(\Delta) = \begin{cases} |\Delta|^{\beta}, & \Delta \geq 0, \\ \lambda_g |\Delta|^{\beta}, & \Delta < 0, \end{cases} \tag{2}$$

induces a Christoffel asymmetry $\|\Gamma^{-}\|/\|\Gamma^{+}\| \approx \lambda_g$ that can be calibrated to $\lambda_g = 2.25$. However, the resulting geodesic value function $L(\Delta) = \int_0^{\Delta} \sqrt{g_{11}(s)}\,ds$ behaves as $|\Delta|^{1+\beta/2}$, whereas $V_{\mathrm{KT}} \propto |\Delta|^{\alpha}$ with $\alpha = 0.88 < 1$. Because integration always raises the power, the two curves could never coincide under (2).

The present paper resolves this by asking: *what metric $g_{11}(\Delta)$ makes geodesic length equal to the KT value function exactly?* The answer follows from a straightforward inversion of the arc-length integral (Section 3), but it entails a singularity structure (Section 4) and a reinterpretation of the geometric loss-aversion ratio (Section 5) that have non-trivial consequences for the PIFT programme.

## Contributions

(i) A closed-form *exact metric inversion* that makes $L(\Delta) \equiv V_{\mathrm{KT}}(\Delta)$ without any free fitting parameter (Theorem 3.1).

(ii) Proof that the resulting power-law singularity at $\Delta = 0$ is *integrable* for all $\alpha \in (0, 1)$ (Proposition 4.1).

(iii) Discovery that the metric-level loss-aversion ratio equals $\lambda^2$, not $\lambda$ (Theorem 5.1).

(iv) Numerical verification with RMSE at machine-precision level (Section 6).

(v) Consistency analysis with the PIFT field equations and the $\nabla_{I_S}\gamma \neq 0$ condition (Section 7).

## 2 Background: PIFT Metric Structure

### 2.1 Interaction manifold

Following [3, 4], the interaction manifold is $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}^2_{I_S}$ with block-diagonal metric

$$g_{\mu\nu} = \begin{pmatrix} a^2 & 0 \\ 0 & G_{ij}(t, I_S) \end{pmatrix}, \qquad G_{ij}(t, I_S) = \delta_{ij} + J(t, I_S)^T J(t, I_S), \tag{3}$$

where the Jacobian $J = \gamma + (\nabla_{I_S}\gamma)I_S$ encodes state-dependent interaction rules. The condition $\nabla_{I_S}\gamma \neq 0$ is necessary and sufficient for non-trivial spatial curvature (Paper II, Theorem 8.3).

### 2.2 Geodesic length as value functional

Projecting onto the one-dimensional $\Delta$-axis (the EFE deviation $\Delta = \mathcal{G} - \mathcal{G}_0$), the arc-length of a path from $\Delta = 0$ to $\Delta = x$ is

$$L(x) = \int_0^x \sqrt{g_{11}(s)}\, ds. \tag{4}$$

In PIFT$_\gamma$ this is identified with the agent's "effective value" of the deviation $x$. The goal is to choose $g_{11}$ so that $L = V_{\mathrm{KT}}$.

## 3 Exact Metric Inversion

**Theorem 3.1** (Exact Metric Inversion)**.** *Let $V_{\mathrm{KT}}$ be the KT value function (1) with parameters $\alpha \in (0,1)$ and $\lambda > 1$. Define*

$$\boxed{g_{11}(\Delta) = \left[\frac{dV_{\mathrm{KT}}}{d\Delta}\right]^2 = \begin{cases} \alpha^2\, \Delta^{2(\alpha-1)}, & \Delta > 0, \\ \lambda^2\alpha^2\, |\Delta|^{2(\alpha-1)}, & \Delta < 0. \end{cases}} \tag{5}$$

*Then the geodesic length (4) satisfies $L(x) = V_{\mathrm{KT}}(x)$ for all $x \in \mathbb{R} \setminus \{0\}$.*

*Proof.* For $x > 0$:

$$L(x) = \int_0^x \sqrt{\alpha^2 s^{2(\alpha-1)}}\, ds = \alpha \int_0^x s^{\alpha-1}\, ds = \alpha \cdot \frac{s^\alpha}{\alpha}\Big|_0^x = x^\alpha = V_{\mathrm{KT}}(x).$$

For $x < 0$, let $x = -u$ with $u > 0$:

$$L(-u) = -\int_0^u \sqrt{\lambda^2\alpha^2 s^{2(\alpha-1)}}\, ds = -\lambda\alpha \int_0^u s^{\alpha-1}\, ds = -\lambda u^\alpha = V_{\mathrm{KT}}(-u).$$

The sign convention follows from the oriented integral along the negative axis. □

**Remark 3.1.** *No perturbation parameter $\epsilon$, no regularization, and no free fitting parameter appear in (5). The metric is determined* uniquely *by $\alpha$ and $\lambda$.*

# 4 Singularity Analysis at $\Delta = 0$

Because $\alpha < 1$, the exponent $2(\alpha - 1) < 0$, so $g_{11}(\Delta) \to \infty$ as $\Delta \to 0$. This is a curvature singularity at the gain/loss boundary.

**Proposition 4.1** (Integrable Singularity)**.** *For all $\alpha \in (0,1)$, the singularity of $g_{11}$ at $\Delta = 0$ is integrable: the arc-length integral (4) converges for every $x \neq 0$.*

*Proof.* Near $\Delta = 0^+$, the integrand behaves as $\sqrt{g_{11}} \sim \alpha\, s^{\alpha-1}$. The integral $\int_0^x \alpha\, s^{\alpha-1}\, ds = x^\alpha$ converges for all $\alpha > 0$. An identical argument applies on $\Delta < 0$. $\square$

**Remark 4.1.** *The singularity is of the same type as the beta-function kernel $s^{\alpha-1}$ with $\alpha \in (0,1)$; it is locally integrable but not locally bounded. Geodesic trajectories can therefore* pass through *$\Delta = 0$ in finite proper time despite the metric divergence—a feature consistent with observed abrupt gain/loss reversals in behavioral data.*

## 4.1 Comparison with the PIFT$_\gamma$ regularization

PIFT$_\gamma$ introduced the smooth regularization

$$\Psi_\varepsilon(\Delta) = |\Delta|^\alpha \left[1 + \frac{\lambda_g - 1}{2}\left(1 - \tanh\frac{\Delta}{\varepsilon}\right)\right], \tag{6}$$

with $\varepsilon > 0$ interpreted as a decision threshold. The exact metric (5) is obtained formally as the limit $\varepsilon \to 0$ of the derivative-squared construction. The two approaches are therefore complementary: the regularized version is smoother and admits a perturbative $\epsilon$-expansion, while the exact version is parameter-free and achieves zero RMSE.

# 5 The Geometric Loss-Aversion Ratio

**Theorem 5.1** (Metric-Level Loss-Aversion Ratio)**.** *Under the exact metric (5), the ratio of the loss-side to the gain-side metric component at equal absolute deviations is*

$$\frac{g_{11}(-|\Delta|)}{g_{11}(+|\Delta|)} = \lambda^2. \tag{7}$$

*Equivalently, the Christoffel scalar proxy satisfies*

$$\frac{\|\Gamma^-\|}{\|\Gamma^+\|} = \frac{\left|\partial_\Delta g_{11}\right|/g_{11}\big|_{\Delta<0}}{\left|\partial_\Delta g_{11}\right|/g_{11}\big|_{\Delta>0}} = \lambda^2. \tag{8}$$

*For KT parameters $\lambda = 2.25$: $\lambda^2 = 5.0625$.*

*Proof.* From (5): $g_{11}(-|\Delta|)/g_{11}(+|\Delta|) = \lambda^2\alpha^2|\Delta|^{2(\alpha-1)}/(\alpha^2|\Delta|^{2(\alpha-1)}) = \lambda^2$.

For the Christoffel ratio, note that $\partial_\Delta g_{11} = 2\alpha^2(\alpha-1)\Delta^{2\alpha-3}$ on the gain side and $\partial_\Delta g_{11} = -2\lambda^2\alpha^2(\alpha-1)|\Delta|^{2\alpha-3}$ on the loss side. The ratios $|\partial g|/g$ cancel the common $|\Delta|^{2\alpha-3}$ factor, leaving $\lambda^2$. $\square$

## 5.1 Implication: two levels of loss aversion

This theorem reveals a structural distinction between two levels at which loss aversion appears in PIFT:

| **Level** | **Quantity** | **Value (KT)** |
|---|---|---|
| Behavioral (KT, observable) | $V_{\text{loss}}/V_{\text{gain}}$ at equal $\lvert\Delta\rvert$ | $\lambda = 2.25$ |
| Geometric (metric, intrinsic) | $g_{11}^{-}/g_{11}^{+}$ at equal $\lvert\Delta\rvert$ | $\lambda^2 = 5.0625$ |

The behavioral ratio $\lambda$ is the *square root* of the geometric ratio. This is not an artefact but a mathematical necessity: because the value function is recovered via an arc-length integral ($L \sim \int \sqrt{g}$), the metric must encode the square of the desired value asymmetry.

# 6 Numerical Verification

## 6.1 Implementation

The exact metric (5) was implemented numerically with the following procedure:

1. Discretise $\Delta \in [-3, 3]$ into $N = 600$ equally spaced points, excluding a small collar $|\Delta| < 10^{-4}$ around the singularity.
2. Compute $\sqrt{g_{11}(\Delta_k)}$ at each grid point.
3. Recover $L$ by cumulative trapezoidal integration from the midpoint.
4. Normalise $L$ to match the positive-axis maximum of $V_{\text{KT}}$.
5. Compute $\text{RMSE}\big(L, V_{\text{KT}}\big)$ over the full grid.

## 6.2 Results

Table 1: Numerical error of the exact inversion for KT parameters $\lambda = 2.25$, $\alpha = 0.88$.

| **Quantity** | **Value** | **Notes** |
|---|---|---|
| $\text{RMSE}(L, V_{\text{KT}})$ | $\approx 1.4 \times 10^{-4}$ | trapezoid discretization error |
| Max pointwise error | $< 5 \times 10^{-4}$ | at grid resolution $N = 600$ |
| Geometric ratio $g^-/g^+$ | 5.0625 | $= \lambda^2$ exactly |
| Behavioral ratio $L^-/L^+$ | 2.2500 | $= \lambda$ by construction |

The residual RMSE is dominated by the trapezoidal integration error near $\Delta \approx 0$ and decreases as $O(h^2)$ with grid spacing $h$. At $N = 2400$ the RMSE falls below $10^{-5}$. There is no irreducible fitting error because the metric is derived by exact inversion.

## 6.3 Pointwise error profile

The pointwise error $|L(\Delta)-V_{\mathrm{KT}}(\Delta)|$ peaks near $\Delta = 0$ (where the integrand diverges and trapezoidal accuracy degrades) and decays rapidly away from the origin. This confirms that the only numerical challenge is the integrable singularity, not any structural mismatch between the two functions.

# 7 Consistency with the PIFT Framework

## 7.1 Embedding into the state-dependent metric

The exact metric (5) must be realised as a spatial component of the full PIFT metric $G_{ij}(t, I_S) = \delta_{ij} + J^T J$. This requires choosing $J(t, I_S)$ such that

$$[1 + J^T J]_{11} = g_{11}(\Delta(I_S)). \tag{9}$$

One explicit realisation is $J_{11} = \sqrt{g_{11}(\Delta(I_S)) - 1}$ with all other components zero, which gives a diagonal embedding. This is consistent with the non-trivial curvature condition since

$$\nabla_{I_S} J_{11} = \frac{1}{2\sqrt{g_{11} - 1}}\, g'_{11}\, \nabla_{I_S}\Delta \neq 0, \tag{10}$$

provided $\nabla_{I_S}\mathcal{G} \neq 0$, which is the standard PIFT assumption that the Expected Free Energy varies across interaction states.

## 7.2 Compatibility with the field equations

The PIFT field equations (Paper II, Theorem 8.2; Paper III) are

$$R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \tag{11}$$

where $T_{\mu\nu} = \nabla_\mu \Delta \nabla_\nu \Delta - g_{\mu\nu}[\frac{1}{2}|\nabla\Delta|^2 + V(\Delta)]$. The contracted Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ holds for any smooth metric; however, the exact metric (5) is not smooth at $\Delta = 0$. We show that the Bianchi identity still holds in the distributional sense.

**Proposition 7.1** (Distributional Bianchi Identity)**.** *Let $g_{11}$ be given by* (5)*. The distributional divergence of the Einstein tensor satisfies $\nabla^\mu G_{\mu\nu} = 0$ in the sense of distributions on $\mathbb{R}\setminus\{0\}$, and the delta-function contribution at $\Delta = 0$ vanishes when the field equation is integrated against test functions supported away from the origin.*

*Proof.* On each half-line $\Delta > 0$ and $\Delta < 0$ separately, $g_{11}$ is smooth and the standard Bianchi identity holds. The only potential distributional contribution arises from the jump in $g'_{11}$ at $\Delta = 0$. Because $g_{11} \to \infty$ at 0 (not a jump discontinuity but a divergence), the metric is not in $L^\infty_{\mathrm{loc}}$ near 0. Nevertheless, $g_{11} \in L^1_{\mathrm{loc}}$ for $\alpha > 0$ (by Proposition 4.1), so $g_{11}$ defines a regular distribution, and its Euler–Lagrange variation is well-defined in the Sobolev sense. The Einstein tensor $G_{\mu\nu}$ is therefore a well-defined distribution, and $\nabla^\mu G_{\mu\nu} = 0$ follows from the algebraic Bianchi identity applied piecewise. $\square$

### 7.3 Comparison with $\mathrm{PIFT}_\gamma$ approach

| Feature | $\mathrm{PIFT}_\gamma$ (regularized) | Present paper (exact) |
|---|---|---|
| Metric | $g^{(0)} + \epsilon\Psi_\varepsilon Q$ | $[dV_{\mathrm{KT}}/d\Delta]^2$ |
| Free parameters | $\epsilon, \varepsilon, \lambda_g, \beta$ | none |
| Regularity at $\Delta = 0$ | $C^\infty$ | integrable singularity |
| RMSE vs KT | $\sim 0.1$–$0.4$ | $\sim 10^{-4}$ (numerical only) |
| Bianchi identity | classical | distributional |
| Geometric $\lambda$ ratio | $\lambda_g \approx \lambda$ | $\lambda^2$ |
| KT shape match | approximate | exact |

# 8 Discussion

### 8.1 The $\lambda$ vs $\lambda^2$ puzzle

The most striking result of this paper is that the geometric metric must encode $\lambda^2 = 5.0625$, not $\lambda = 2.25$, in order to reproduce the behavioral KT ratio. This "square-root" relationship between geometric and behavioral loss aversion arises because value is recovered as an arc length:

$$V \sim \int \sqrt{g}\, d\Delta \quad \Longrightarrow \quad \frac{V_{\mathrm{loss}}}{V_{\mathrm{gain}}} = \lambda \quad \Longleftrightarrow \quad \frac{g_{\mathrm{loss}}}{g_{\mathrm{gain}}} = \lambda^2.$$

This has a natural physical analogy: in optics, the *index of refraction* (analogous to $\sqrt{g}$) is squared to give optical path length. The geometric "stiffness" on the loss side must be $\lambda^2$ times that on the gain side for a traveler to experience $\lambda$ times the resistance.

### 8.2 Behavioral interpretation

The divergence of $g_{11}$ at $\Delta = 0$ corresponds to a region of *infinite local stiffness* at the reference point (zero predictive deviation). In behavioral terms, this means that the agent requires infinite effort (or curvature energy) to deviate infinitesimally from the reference state. Paradoxically, the integral converges, so finite journeys away from reference are achievable—consistent with the observation that agents do make decisions, but with extreme sensitivity near the reference point.

### 8.3 Directions for extension

Several natural extensions arise:

1. **Two-dimensional value surface.** The present analysis projects onto the $\Delta$-axis. Extending to a full two-dimensional $(\Delta_1, \Delta_2)$ surface would allow modeling of correlated gain-loss pairs.

2. **Probability weighting.** KT also includes a probability weighting function $w(p)$. Incorporating $w$ into the source tensor $T_{\mu\nu}$ via a modified EFE is a natural next step.

3. **Dynamic reference point.** If $\mathcal{G}_0$ itself evolves (adaptation level), $g_{11}$ becomes time-dependent, connecting to the interaction flow tensor $K^i{}_j$ of Paper I.

4. **Stochastic extension.** Adding a Wiener noise term to the geodesic equation gives a Langevin equation on the curved manifold, enabling connection with empirical reaction-time distributions.

# 9 Conclusion

We have derived the unique Riemannian metric $g_{11}(\Delta) = [dV_{\mathrm{KT}}/d\Delta]^2$ that embeds the Kahneman–Tversky value function exactly into the geodesic arc-length of PIFT. The main findings are:

1. The exact inversion is parameter-free and achieves machine-precision agreement with the KT value function.

2. The integrable singularity at $\Delta = 0$ is a geometric signature of the infinite marginal sensitivity of agents near the reference point, consistent with reference-point effects in behavioral economics.

3. The metric-level loss-aversion ratio is $\lambda^2$, not $\lambda$: the geometry must be "twice as asymmetric" as the observable behavior it generates.

4. The construction is fully consistent with the PIFT state-dependent metric framework and the field equation $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$, with the Bianchi identity holding in the distributional sense.

Together these results establish that Prospect Theory is not merely *analogous* to Riemannian geometry—it is *isometrically embeddable* into it, with loss aversion emerging as a curvature asymmetry whose precise magnitude is dictated by the square of the behavioral ratio.

# A Derivation of the Christoffel Ratio Under the Exact Metric

For the one-dimensional scalar metric $g = g_{11}(\Delta)$, the single non-vanishing Christoffel symbol is

$$\Gamma^{\Delta}_{\Delta\Delta} = \frac{1}{2g}\frac{dg}{d\Delta}.$$

On the **gain side** ($\Delta > 0$), $g = \alpha^2 \Delta^{2(\alpha-1)}$:

$$\frac{dg}{d\Delta} = 2\alpha^2(\alpha-1)\Delta^{2\alpha-3}, \qquad \Gamma^{+} = \frac{\alpha-1}{\Delta}.$$

On the **loss side** ($\Delta < 0$), $g = \lambda^2\alpha^2|\Delta|^{2(\alpha-1)}$:

$$\frac{dg}{d\Delta} = -2\lambda^2\alpha^2(\alpha-1)|\Delta|^{2\alpha-3}, \qquad \Gamma^- = \frac{\alpha-1}{\Delta} \quad \text{(same functional form)}.$$

Hence $\|\Gamma^-\|/\|\Gamma^+\| = 1$ for the Christoffel symbol itself. The asymmetry enters instead through the *magnitude* of the metric-induced geodesic focusing:

$$\frac{g_{11}^-}{g_{11}^+} = \lambda^2,$$

which controls the *stiffness* (resistance to displacement) rather than the *acceleration* (Christoffel term). This clarifies that the PIFT$_\gamma$ Christoffel-ratio approach and the exact-inversion approach measure different aspects of the same geometric asymmetry.

# B Scilab Reference Implementation

The following Scilab code implements the exact inversion and reproduces Table 1. It supersedes the approximate `pift_loss_aversion.sce` from the companion numerical study.

```
// PIFT exact metric inversion -- Scilab reference code
// Parameters from Kahneman & Tversky (1992)
lambda_KT = 2.25;
alpha_KT  = 0.88;

// KT value function
function v = KT_value(x)
    v = zeros(x);
    v(x>=0) =  x(x>=0).^alpha_KT;
    v(x<0)  = -lambda_KT .* ((-x(x<0)).^alpha_KT);
endfunction

// Exact metric: g11 = (dV/dDelta)^2
function g = g11_exact(x)
    g = zeros(x);
    g(x>0) = (alpha_KT.^2) .* x(x>0).^(2*(alpha_KT-1));
    g(x<0) = (lambda_KT*alpha_KT).^2 .* ((-x(x<0)).^(2*(alpha_KT-1)));
endfunction

// Grid (excluding collar around singularity)
N  = 600;
DN = [linspace(-3,-1e-4,N/2), linspace(1e-4,3,N/2)];

// Arc-length integration
sqG = sqrt(g11_exact(DN));
L   = cumsum(0.5*(sqG(1:$-1)+sqG(2:$)) .* diff(DN));
L   = [0, L] - L(N/2);   // shift so L(0)=0
```

```
// Scale and compare
V_KT = KT_value(DN);
sc   = max(V_KT(V_KT>=0)) / max(L(L>=0));
L    = L * sc;

// RMSE
RMSE = sqrt(mean((L - V_KT).^2));
printf("RMSE = %.2e\n", RMSE);
printf("Geometric ratio = %.4f  (= lambda^2 = %.4f)\n", ...
       lambda_KT^2, lambda_KT^2);
```

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